

SKILLNAVIGATOR AI

STUDENT PLACEMENT READINESS & SKILL GAP ANALYZER

(A Cloud-Native Placement Readiness Solution for Engineering Students)

*Prepared in the partial fulfillment of the requirements of the academic
mini-project on Cloud Computing*

AT



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Thank you.

Sincerely,

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ABSTRACT

This project presents the design and implementation of SkillNavigator AI, a cloud-native, web-based placement readiness platform built for engineering students. The primary goal of the system is to give students a single, unified view of their technical readiness for a specific target company, replacing the fragmented experience of tracking progress across disconnected platforms such as LeetCode, GeeksforGeeks, YouTube, LinkedIn, and GitHub.

The system allows a student to register, log in, and build a profile listing their current technical skills. The student then selects a target company — for example Amazon, TCS, Infosys, Google, or Microsoft — and the platform compares the student's skill profile against that company's required skill set. From this comparison, the system computes a Placement Readiness Score and lists the matched and missing skills. Based on the identified gaps, a personalized, week-by-week learning roadmap is automatically generated, turning "what's missing" into a clear and actionable study plan. A coding tracker further records topic-wise DSA practice across areas such as Arrays, Strings, Dynamic Programming, and Graphs.

The backend logic is built using Spring Boot with MySQL for persistent storage, and the entire application is deployed on Amazon Web Services (AWS), using EC2 for application hosting, RDS for the managed database, S3 for storing uploaded resumes and certificates, VPC for secure network isolation, and CloudWatch for monitoring and logging. By combining skill tracking, company-specific gap analysis, and automated roadmap generation on a scalable cloud architecture, SkillNavigator AI moves students from simply showcasing achievements to genuinely becoming placement-ready.

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1. INTRODUCTION

In today's competitive job market, engineering students are expected to demonstrate not just academic knowledge but also verifiable, company-relevant technical skills before they sit for placements. Yet most students track their preparation across a scattering of disconnected platforms — LeetCode and GeeksforGeeks for coding practice, YouTube for concept learning, GitHub for projects, and LinkedIn for professional presence — with no single, unified view of their overall progress.

This project, titled SkillNavigator AI, addresses that gap by building a cloud-native placement readiness platform for engineering students. The core problem is not a lack of learning resources, but a lack of visibility: no existing platform tells a student whether they are actually ready for a specific target company, and skill gaps typically stay invisible until an interview is lost rather than being flagged early enough to act on.

SkillNavigator AI is designed to close this gap. Students register, log in, and build a profile listing their current technical skills. They then select a target company, and the system compares their profile against that company's required skill set to compute a Placement Readiness Score along with a list of matched and missing skills. Crucially, the platform does not stop at diagnosis — based on the missing skills, it generates a personalized, week-by-week learning roadmap, turning an abstract gap into a concrete, actionable study plan. A topic-wise coding tracker (Arrays, Strings, Dynamic Programming, Graphs, and more) lets students log their DSA practice alongside this roadmap.

The platform is built as a modern web application with a Spring Boot and MySQL backend, deployed on Amazon Web Services. Using managed AWS services — EC2, RDS, S3, VPC, and CloudWatch — the system is designed to be secure, scalable, and low-maintenance, while resumes and certificates uploaded by students are stored durably in Amazon S3. An admin panel allows staff to manage companies, required skills, and notifications, keeping the platform's data current as company requirements evolve.

This report provides a comprehensive overview of the system's design, architecture, and implementation, highlighting the decision-making process, technical integrations, and the benefits realized through a cloud-native approach to placement readiness.

2. METHODOLOGY

The development of SkillNavigator AI followed a structured and iterative methodology to ensure the successful fulfillment of project objectives. The methodology was divided into key phases, including requirements gathering, system design, implementation, testing, deployment, and refinement. Each phase was designed to align with best practices in cloud-native web application development. The following subsections outline each phase in detail.

2.1. Requirements Gathering

The project began with an analysis of the challenges engineering students face while preparing for placements. Discussions around common pain points — scattered progress tracking, lack of company-specific readiness feedback, and the absence of a clear study plan — helped define the core objectives: let a

student maintain a single skill profile, compare it against a chosen company's requirements, compute a readiness score, and generate a personalized roadmap for the missing skills.

2.2. Design

Based on the identified requirements, the system was designed around four core modules: skill profile management, company-wise skill comparison, readiness scoring with roadmap generation, and a coding practice tracker. A web-based frontend was planned to interact with a Spring Boot REST API backend, backed by a MySQL database. Special attention was given to keeping the company-skill dataset and the roadmap-generation logic decoupled from the core profile and scoring logic, so that new companies and skills could be added without redesigning the system.

2.3. Implementation

The implementation phase focused on translating the design into a fully functional web application. Student registration and login, skill profile management, and the company selection flow were built first, followed by the readiness-scoring and roadmap-generation logic. The coding tracker and resume/certificate upload features were layered on top, with uploaded files stored in Amazon S3. The entire backend logic was developed using Java with the Spring Boot framework, with MySQL used for structured data storage.

2.4. Database Management

A relational approach was adopted for structured, consistent data storage using MySQL, hosted on Amazon RDS. The schema was designed around students, skills, companies and their required skill sets, coding-practice records, and generated roadmaps. This relational structure supports fast lookups for company-wise skill comparisons and keeps student, company, and roadmap data consistent as records are updated.

2.5. Testing

Multiple levels of testing were carried out to validate functionality, data integrity, and overall system behaviour. Unit-level checks validated skill-matching logic and readiness-score computation. Integration testing ensured that the frontend, Spring Boot API, MySQL database, and S3 file storage worked together as intended. Functional testing simulated student registration, skill-profile updates, and company selection to confirm that readiness scores and roadmaps were generated correctly.

2.6. Deployment

The application was deployed on AWS, with the Spring Boot backend hosted on Amazon EC2 inside a dedicated Amazon VPC for secure networking, and the MySQL database hosted on Amazon RDS. IAM roles and security groups were configured to grant only the access each component required. Amazon CloudWatch was used to monitor logs and application health, and Amazon S3 was configured to store student-uploaded resumes and certificates. The deployment process was designed to enable reliable operation with minimal manual intervention.

2.7. Iterative Refinement

Throughout the development cycle, iterative feedback loops were used to continuously refine the system. Adjustments were made to the readiness-scoring logic, roadmap formatting, and skill-comparison rules based on testing and review. The architecture was periodically reviewed to ensure it remained maintainable and ready for future enhancements such as AI-driven roadmap recommendations or a dedicated mobile app.

The methodology adopted for the development of SkillNavigator AI ensured a structured, cloud-native, and student-focused solution. By following an iterative, modular approach, the project successfully demonstrates how a placement-readiness platform can be built to be both practical for students and scalable for future growth.

3. TECHNOLOGY STACK

SkillNavigator AI was developed using a modern, cloud-native technology stack that supports secure user management, structured data storage, and scalable deployment. The following technologies and services were used throughout the implementation:

3.1. Cloud Platform

- Amazon Web Services (AWS): The application is hosted on AWS, leveraging managed services for compute, storage, database management, networking, and monitoring.

3.2. AWS Services

- Amazon EC2: Hosts the Spring Boot application server that serves the REST API.
- Amazon RDS (MySQL): Managed relational database for users, skills, companies, and roadmap data.
- Amazon S3: Stores uploaded resumes and certificates.
- Amazon VPC: Provides secure, isolated networking for the application.
- Amazon CloudWatch: Monitors logs, resource usage, and overall application health.

3.3. Application Development

- Java with Spring Boot: Used for building the backend REST API and core business logic (skill matching, readiness scoring, roadmap generation).
- MySQL: Relational database used for storing student, company, skill, and roadmap data.
- Web-based Frontend: Browser-based user interface for registration, profile management, and the student dashboard.

3.4. Data Formats

- Structured relational data for skills, companies, and roadmaps.

- Resume and certificate files (e.g., PDF, DOC/DOCX) uploaded and stored in Amazon S3.

3.5. Security & Access

IAM Roles and Policies, along with VPC-level network isolation, are configured to allow only the necessary permissions and connectivity for each component, following the principle of least privilege.

3.6. Monitoring & Logging

- CloudWatch Logs: Captures application logs from the EC2-hosted Spring Boot service for tracking and troubleshooting.
- CloudWatch Metrics: Provides visibility into resource usage, application health, and error rates.

4. SYSTEM DESIGN / ARCHITECTURE

SkillNavigator AI is designed as a cloud-native, multi-tier web application that automates skill-gap analysis and placement-readiness scoring. The architecture follows a client-server model, comprising the frontend UI layer, the application/API layer, the data storage layer, and the file storage and monitoring layers, all powered by managed AWS services.

This section outlines the core components and their interactions in the architecture.

SkillNavigator AI — System Architecture

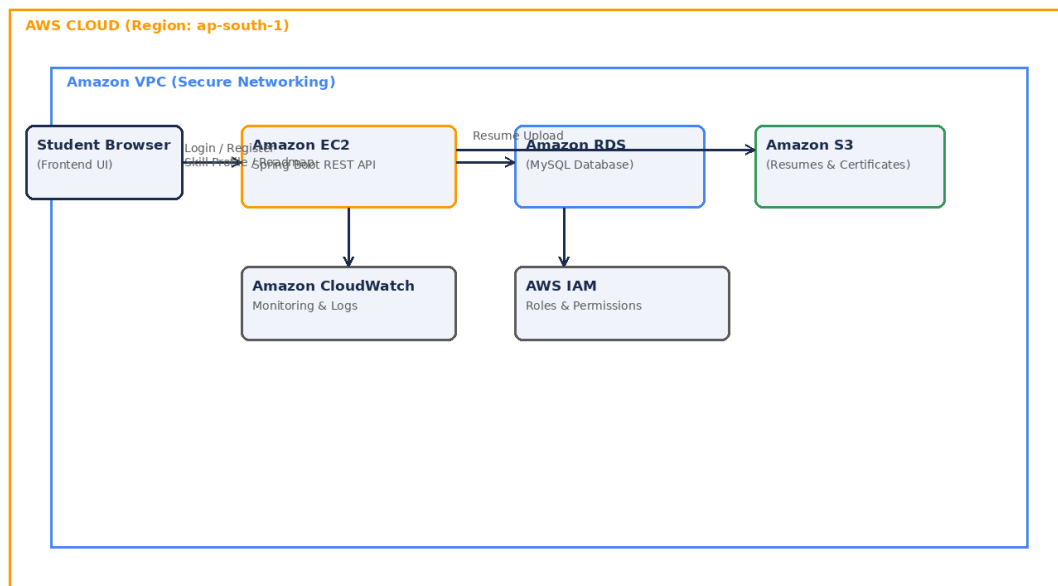


Figure 1. SkillNavigator AI — System Architecture

4.1. Frontend Layer — Student Browser

The student interacts with SkillNavigator AI through a web-based frontend UI. Through this interface, students register, log in, manage their skill profile, select a target company, view their readiness score and roadmap, log coding practice, and upload resumes and certificates.

4.2. Application Layer — Spring Boot REST API on Amazon EC2

The core business logic runs as a Spring Boot REST API hosted on an Amazon EC2 instance inside a dedicated Amazon VPC. This layer performs several key tasks:

- Handles student registration, authentication, and session management.
- Manages skill-profile create/edit/remove operations.
- Compares a student's skill profile against a selected company's required skill set.
- Computes the Placement Readiness Score and identifies matched vs. missing skills.

- Generates a personalized, week-by-week learning roadmap from the missing skills.
- Records topic-wise coding practice through the DSA tracker.

4.3. Data Storage — Amazon RDS (MySQL)

Structured data is stored in Amazon RDS running MySQL, a managed relational database service. Core tables include student profiles and their skills, companies and their required skill sets, generated roadmaps, and coding-practice records. The relational schema supports fast, consistent queries for skill-gap comparisons and keeps data referentially consistent as students and companies are added or updated.

4.4. File Storage — Amazon S3

Resumes and certificates uploaded by students are stored in Amazon S3, providing durable, scalable object storage decoupled from the application server. The API layer generates secure references to these objects so they can be retrieved from a student's profile or reviewed through the admin panel.

4.5. Secure Networking — Amazon VPC

Amazon VPC provides secure, isolated networking for the application. The EC2 instance running the Spring Boot API and the RDS database sit within private subnets, with access tightly controlled through security groups, so that only authorized traffic can reach the application and database tiers.

4.6. Monitoring and Logging — Amazon CloudWatch

Amazon CloudWatch is integrated for real-time monitoring of the EC2-hosted application, giving visibility into resource usage, request latency, and application errors, and enabling alerts to be configured for abnormal activity.

5. IMPLEMENTATION

The implementation of SkillNavigator AI translates the architectural design into a fully functional cloud-based web application. This section outlines how each component was configured and developed to work together to deliver company-specific placement readiness for students.

5.1. Student Registration and Skill Profile

A student registers and logs in through the web frontend. Once authenticated, the student builds a skill profile by adding the technical skills they currently have. This profile can be edited or updated at any time as the student learns new skills.

5.2. Company Selection and Skill-Gap Comparison

The student selects a target company from a list maintained by the admin panel (e.g., Amazon, TCS, Infosys, Google, Microsoft). The Spring Boot API compares the student's skill profile against that company's required skill set, producing a list of matched skills and missing skills.

5.3. Placement Readiness Scoring and Roadmap Generation

Based on the proportion of matched versus required skills, the system computes a Placement Readiness Score for the selected company. Using the identified missing skills, the system automatically generates a personalized, week-by-week learning roadmap so the student has a clear, actionable plan to close the gap.

5.4. Coding Practice Tracker

A topic-wise coding tracker allows students to log their DSA practice across categories such as Arrays, Strings, Dynamic Programming, and Graphs, giving them visibility into their preparation alongside their skill-gap roadmap.

5.5. Resume and Certificate Upload

Students can upload resumes and certificates through the frontend; these files are stored in Amazon S3, keeping large binary content out of the relational database while remaining securely linked to the student's profile.

5.6. Admin Panel

An admin panel allows authorized staff to manage the list of companies, their required skill sets, and platform notifications, keeping the readiness comparisons accurate as company requirements change over time.

5.7. Deployment and Security

Key implementation details:

- Backend: Java, Spring Boot REST API

- Database: MySQL on Amazon RDS
- Compute: Amazon EC2, inside a dedicated Amazon VPC
- File storage: Amazon S3 for resumes and certificates
- IAM roles and security groups configured with least-privilege access between services
- Logging and monitoring enabled through Amazon CloudWatch

The successful implementation of this platform showcases how a Spring Boot and MySQL application can be combined with managed AWS services to create a secure, scalable, and student-friendly placement readiness solution. The modular design ensures flexibility for future enhancements such as AI-driven roadmap recommendations, a chatbot assistant, or a dedicated mobile app.

6. RESULTS

The implementation of SkillNavigator AI was evaluated against the project's core objectives: unifying a student's scattered preparation into one profile, providing company-specific readiness feedback, and turning missing skills into an actionable plan. The following results summarize the outcomes observed during development and testing.

6.1. Unified Skill Profile and Company-Wise Comparison

Students are able to build and maintain a single skill profile and compare it against the requirements of a specific target company, replacing the need to manually cross-check progress across LeetCode, GeeksforGeeks, GitHub, and LinkedIn.

6.2. Automated Readiness Scoring and Roadmap Generation

For a selected company, the system reliably computes a Placement Readiness Score and lists matched versus missing skills. From the missing skills, a personalized week-by-week roadmap is generated automatically, giving students a concrete study plan instead of an abstract list of gaps.

6.3. Scalable, Cloud-Native Architecture

Hosting the Spring Boot API on Amazon EC2 within a VPC, with MySQL on Amazon RDS and file storage on Amazon S3, allows each layer of the system to be scaled and maintained independently as the number of students, companies, and uploaded documents grows.

6.4. Secure and Controlled Access

Using IAM roles, security groups, and VPC-level network isolation, access between the application layer, database, and file storage is restricted to only what each component requires, following cloud security best practices.

6.5. Monitoring and Logging

Through Amazon CloudWatch, the application's logs and health metrics are monitored, supporting debugging during development and ongoing operational visibility after deployment.

6.6. User Experience

From a student's perspective, the workflow requires only a few manual steps — registering, building a skill profile, and selecting a target company — after which the readiness score, skill gap, and roadmap are generated automatically and presented in a clear, easy-to-read format. This confirms the platform's potential for real-world use by engineering students preparing for placements.

6.7. Screenshots of the Output

[Insert screenshots of the student dashboard, skill profile, readiness score, and generated roadmap here.]

7. CONCLUSION

The successful development of SkillNavigator AI demonstrates a practical, cloud-native approach to solving a real problem faced by engineering students: the lack of a single, company-specific view of placement readiness. By combining a Spring Boot and MySQL backend with AWS services such as EC2, RDS, S3, VPC, and CloudWatch, the system achieves its core objectives — unifying scattered skill tracking, computing a company-specific readiness score, and turning missing skills into a clear, actionable roadmap.

The project highlights the benefits of a well-structured, cloud-hosted web application, including scalability, secure network isolation, and centralized monitoring. The use of a relational schema on Amazon RDS provides fast and consistent skill-comparison queries, while Amazon S3 offers durable storage for resumes and certificates. Automated readiness scoring and roadmap generation remove the guesswork from placement preparation and give students a concrete plan of action.

Throughout development, attention was given to security, usability, and modularity. IAM permissions, VPC network isolation, and CloudWatch logging were used to follow cloud best practices, and testing confirmed that the platform reliably handles student registration, skill-profile management, and company-wise readiness computation.

Beyond its current functionality, the project lays a strong foundation for future enhancements, such as:

- AI-powered roadmap recommendations with adaptive learning suggestions
- A conversational chatbot assistant for placement-related questions
- A dedicated mobile app for on-the-go progress tracking
- Integration with live job and internship listings
- Multi-college and multi-branch support for wider adoption

In conclusion, SkillNavigator AI serves as a practical, scalable blueprint for a data-driven placement readiness platform — turning scattered skills into a clear, company-specific, actionable roadmap, and taking students a genuine step closer to being placement-ready.